IDFT notebook - executed PDF export

Discrete-time signal synthesis using the IDFT

Input spectrum: X = [6, 2, 4, 3, 4, 4, 0, 0, 0, 0]^T.

Convention: W[n,k] = exp(-j 2 pi n k / N), K = W^H / N, x = K X.
The exercise does not define the letters W and K; these names are explicitly chosen here.

N = 10 is the supplied case. N = 16 and N = 20 are illustrative additional cases using trailing zero padding of the spectrum. No real-valued constraint or conjugate symmetry is imposed.

Python code

```
import numpy as np
import matplotlib.pyplot as plt

X = np.array([6, 2, 4, 3, 4, 4, 0, 0, 0, 0], dtype=float)
N_values = [10, 16, 20]
results = {}
fig, axes = plt.subplots(3, 1, figsize=(10, 9), constrained_layout=True)
for ax, N in zip(axes, N_values):
    XN = np.pad(X, (0, N - len(X)))
    n = np.arange(N)
    W = np.exp(-2j * np.pi * np.outer(n, n) / N)
    K = W.conj().T / N
    x = K @ XN
    assert np.allclose(K @ W, np.eye(N))
    assert np.allclose(x, np.fft.ifft(XN))
    results[N] = x
    np.set_printoptions(precision=4, linewidth=180)
    print(f"N = {N}\nW =\n{W}\nK =\n{K}\nx =\n{x}\n")
    ax.stem(n, x.real, linefmt='C0-', markerfmt='C0o',
            basefmt='k-', label='Real part')
    ax.stem(n, x.imag, linefmt='C1--', markerfmt='C1x',
            basefmt=' ', label='Imaginary part')
    ax.set(title=f'N = {N}', xlabel='Sample n', ylabel='Amplitude')
    ax.grid(alpha=0.25)
    ax.legend()
plt.show()
```

Executed output: full W, K and synthesized samples

```
N = 10
W =
[[ 1.   +0.0000e+00j  1.   +0.0000e+00j  1.   +0.0000e+00j  1.   +0.0000e+00j  1.   +0.0000e+00j  1.   +0.0000e+00j  1.   +0.0000e+00j  1.   +0.0000e+00j  1.   +0.0000e+00j
   1.   +0.0000e+00j]
 [ 1.   +0.0000e+00j  0.809-5.8779e-01j  0.309-9.5106e-01j -0.309-9.5106e-01j -0.809-5.8779e-01j -1.   -1.2246e-16j -0.809+5.8779e-01j -0.309+9.5106e-01j  0.309+9.5106e-01j
   0.809+5.8779e-01j]
 [ 1.   +0.0000e+00j  0.309-9.5106e-01j -0.809-5.8779e-01j -0.809+5.8779e-01j  0.309+9.5106e-01j  1.   +2.4493e-16j  0.309-9.5106e-01j -0.809-5.8779e-01j -0.809+5.8779e-01j
   0.309+9.5106e-01j]
 [ 1.   +0.0000e+00j -0.309-9.5106e-01j -0.809+5.8779e-01j  0.809+5.8779e-01j  0.309-9.5106e-01j -1.   -3.6739e-16j  0.309+9.5106e-01j  0.809-5.8779e-01j -0.809-5.8779e-01j
  -0.309+9.5106e-01j]
 [ 1.   +0.0000e+00j -0.809-5.8779e-01j  0.309+9.5106e-01j  0.309-9.5106e-01j -0.809+5.8779e-01j  1.   +4.8986e-16j -0.809-5.8779e-01j  0.309+9.5106e-01j  0.309-9.5106e-01j
  -0.809+5.8779e-01j]
 [ 1.   +0.0000e+00j -1.   -1.2246e-16j  1.   +2.4493e-16j -1.   -3.6739e-16j  1.   +4.8986e-16j -1.   +1.1640e-15j  1.   +7.3479e-16j -1.   -8.5725e-16j  1.   +9.7972e-16j
  -1.   +2.4505e-15j]
 [ 1.   +0.0000e+00j -0.809+5.8779e-01j  0.309-9.5106e-01j  0.309+9.5106e-01j -0.809-5.8779e-01j  1.   +7.3479e-16j -0.809+5.8779e-01j  0.309-9.5106e-01j  0.309+9.5106e-01j
  -0.809-5.8779e-01j]
 [ 1.   +0.0000e+00j -0.309+9.5106e-01j -0.809-5.8779e-01j  0.809-5.8779e-01j  0.309+9.5106e-01j -1.   -8.5725e-16j  0.309-9.5106e-01j  0.809+5.8779e-01j -0.809+5.8779e-01j
  -0.309-9.5106e-01j]
 [ 1.   +0.0000e+00j  0.309+9.5106e-01j -0.809+5.8779e-01j -0.809-5.8779e-01j  0.309-9.5106e-01j  1.   +9.7972e-16j  0.309+9.5106e-01j -0.809+5.8779e-01j -0.809-5.8779e-01j
   0.309-9.5106e-01j]
 [ 1.   +0.0000e+00j  0.809+5.8779e-01j  0.309+9.5106e-01j -0.309+9.5106e-01j -0.809+5.8779e-01j -1.   +2.4505e-15j -0.809-5.8779e-01j -0.309-9.5106e-01j  0.309-9.5106e-01j
   0.809-5.8779e-01j]]
K =
[[ 0.1   -0.0000e+00j  0.1   -0.0000e+00j  0.1   -0.0000e+00j  0.1   -0.0000e+00j  0.1   -0.0000e+00j  0.1   -0.0000e+00j  0.1   -0.0000e+00j  0.1   -0.0000e+00j
   0.1   -0.0000e+00j  0.1   -0.0000e+00j]
 [ 0.1   -0.0000e+00j  0.0809+5.8779e-02j  0.0309+9.5106e-02j -0.0309+9.5106e-02j -0.0809+5.8779e-02j -0.1   +1.2246e-17j -0.0809-5.8779e-02j -0.0309-9.5106e-02j
   0.0309-9.5106e-02j  0.0809-5.8779e-02j]
 [ 0.1   -0.0000e+00j  0.0309+9.5106e-02j -0.0809+5.8779e-02j -0.0809-5.8779e-02j  0.0309-9.5106e-02j  0.1   -2.4493e-17j  0.0309+9.5106e-02j -0.0809+5.8779e-02j
  -0.0809-5.8779e-02j  0.0309-9.5106e-02j]
 [ 0.1   -0.0000e+00j -0.0309+9.5106e-02j -0.0809-5.8779e-02j  0.0809-5.8779e-02j  0.0309+9.5106e-02j -0.1   +3.6739e-17j  0.0309-9.5106e-02j  0.0809+5.8779e-02j
  -0.0809+5.8779e-02j -0.0309-9.5106e-02j]
 [ 0.1   -0.0000e+00j -0.0809+5.8779e-02j  0.0309-9.5106e-02j  0.0309+9.5106e-02j -0.0809-5.8779e-02j  0.1   -4.8986e-17j -0.0809+5.8779e-02j  0.0309-9.5106e-02j
   0.0309+9.5106e-02j -0.0809-5.8779e-02j]
 [ 0.1   -0.0000e+00j -0.1   +1.2246e-17j  0.1   -2.4493e-17j -0.1   +3.6739e-17j  0.1   -4.8986e-17j -0.1   -1.1640e-16j  0.1   -7.3479e-17j -0.1   +8.5725e-17j
   0.1   -9.7972e-17j -0.1   -2.4505e-16j]
 [ 0.1   -0.0000e+00j -0.0809-5.8779e-02j  0.0309+9.5106e-02j  0.0309-9.5106e-02j -0.0809+5.8779e-02j  0.1   -7.3479e-17j -0.0809-5.8779e-02j  0.0309+9.5106e-02j
   0.0309-9.5106e-02j -0.0809+5.8779e-02j]
 [ 0.1   -0.0000e+00j -0.0309-9.5106e-02j -0.0809+5.8779e-02j  0.0809+5.8779e-02j  0.0309-9.5106e-02j -0.1   +8.5725e-17j  0.0309+9.5106e-02j  0.0809-5.8779e-02j
  -0.0809-5.8779e-02j -0.0309+9.5106e-02j]
 [ 0.1   -0.0000e+00j  0.0309-9.5106e-02j -0.0809-5.8779e-02j -0.0809+5.8779e-02j  0.0309+9.5106e-02j  0.1   -9.7972e-17j  0.0309-9.5106e-02j -0.0809-5.8779e-02j
  -0.0809+5.8779e-02j  0.0309+9.5106e-02j]
 [ 0.1   -0.0000e+00j  0.0809-5.8779e-02j  0.0309-9.5106e-02j -0.0309-9.5106e-02j -0.0809-5.8779e-02j -0.1   -2.4505e-16j -0.0809+5.8779e-02j -0.0309+9.5106e-02j
   0.0309+9.5106e-02j  0.0809+5.8779e-02j]]
x =
[2.3   +0.0000e+00j 0.0691+1.0184e+00j 0.6191-1.3143e-01j 0.1809+1.5918e-01j 0.7309-2.1266e-01j 0.5   -6.2482e-16j 0.7309+2.1266e-01j 0.1809-1.5918e-01j 0.6191+1.3143e-01j
 0.0691-1.0184e+00j]

N = 16
W =
[[ 1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j
   1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j
   1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j]
 [ 1.0000e+00+0.0000e+00j  9.2388e-01-3.8268e-01j  7.0711e-01-7.0711e-01j  3.8268e-01-9.2388e-01j  6.1232e-17-1.0000e+00j -3.8268e-01-9.2388e-01j -7.0711e-01-7.0711e-01j
```

```
  -9.2388e-01-3.8268e-01j -1.0000e+00-1.2246e-16j -9.2388e-01+3.8268e-01j -7.0711e-01+7.0711e-01j -3.8268e-01+9.2388e-01j -1.8370e-16+1.0000e+00j  3.8268e-01+9.2388e-01j
   7.0711e-01+7.0711e-01j  9.2388e-01+3.8268e-01j]
 [ 1.0000e+00+0.0000e+00j  7.0711e-01-7.0711e-01j  6.1232e-17-1.0000e+00j -7.0711e-01-7.0711e-01j -1.0000e+00-1.2246e-16j -7.0711e-01+7.0711e-01j -1.8370e-16+1.0000e+00j
   7.0711e-01+7.0711e-01j  1.0000e+00+2.4493e-16j  7.0711e-01-7.0711e-01j  3.0616e-16-1.0000e+00j -7.0711e-01-7.0711e-01j -1.0000e+00-3.6739e-16j -7.0711e-01+7.0711e-01j
  -4.2863e-16+1.0000e+00j  7.0711e-01+7.0711e-01j]
 [ 1.0000e+00+0.0000e+00j  3.8268e-01-9.2388e-01j -7.0711e-01-7.0711e-01j -9.2388e-01+3.8268e-01j -1.8370e-16+1.0000e+00j  9.2388e-01+3.8268e-01j  7.0711e-01-7.0711e-01j
  -3.8268e-01-9.2388e-01j -1.0000e+00-3.6739e-16j -3.8268e-01+9.2388e-01j  7.0711e-01+7.0711e-01j  9.2388e-01-3.8268e-01j  5.5109e-16-1.0000e+00j -9.2388e-01-3.8268e-01j
  -7.0711e-01+7.0711e-01j  3.8268e-01+9.2388e-01j]
 [ 1.0000e+00+0.0000e+00j  6.1232e-17-1.0000e+00j -1.0000e+00-1.2246e-16j -1.8370e-16+1.0000e+00j  1.0000e+00+2.4493e-16j  3.0616e-16-1.0000e+00j -1.0000e+00-3.6739e-16j
  -4.2863e-16+1.0000e+00j  1.0000e+00+4.8986e-16j  5.5109e-16-1.0000e+00j -1.0000e+00-6.1232e-16j -2.4499e-15+1.0000e+00j  1.0000e+00+7.3479e-16j -9.8034e-16-1.0000e+00j
  -1.0000e+00-8.5725e-16j -2.6948e-15+1.0000e+00j]
 [ 1.0000e+00+0.0000e+00j -3.8268e-01-9.2388e-01j -7.0711e-01+7.0711e-01j  9.2388e-01+3.8268e-01j  3.0616e-16-1.0000e+00j -9.2388e-01+3.8268e-01j  7.0711e-01+7.0711e-01j
   3.8268e-01-9.2388e-01j -1.0000e+00-6.1232e-16j  3.8268e-01+9.2388e-01j  7.0711e-01-7.0711e-01j -9.2388e-01-3.8268e-01j -2.6948e-15+1.0000e+00j  9.2388e-01-3.8268e-01j
  -7.0711e-01-7.0711e-01j -3.8268e-01+9.2388e-01j]
 [ 1.0000e+00+0.0000e+00j -7.0711e-01-7.0711e-01j -1.8370e-16+1.0000e+00j  7.0711e-01-7.0711e-01j -1.0000e+00-3.6739e-16j  7.0711e-01+7.0711e-01j  5.5109e-16-1.0000e+00j
  -7.0711e-01+7.0711e-01j  1.0000e+00+7.3479e-16j -7.0711e-01-7.0711e-01j -2.6948e-15+1.0000e+00j  7.0711e-01-7.0711e-01j -1.0000e+00-1.1022e-15j  7.0711e-01+7.0711e-01j
  -4.9048e-16-1.0000e+00j -7.0711e-01+7.0711e-01j]
 [ 1.0000e+00+0.0000e+00j -9.2388e-01-3.8268e-01j  7.0711e-01+7.0711e-01j -3.8268e-01-9.2388e-01j -4.2863e-16+1.0000e+00j  3.8268e-01-9.2388e-01j -7.0711e-01+7.0711e-01j
   9.2388e-01-3.8268e-01j -1.0000e+00-8.5725e-16j  9.2388e-01+3.8268e-01j -7.0711e-01-7.0711e-01j  3.8268e-01+9.2388e-01j -4.9048e-16-1.0000e+00j -3.8268e-01+9.2388e-01j
   7.0711e-01-7.0711e-01j -9.2388e-01+3.8268e-01j]
 [ 1.0000e+00+0.0000e+00j -1.0000e+00-1.2246e-16j  1.0000e+00+2.4493e-16j -1.0000e+00-3.6739e-16j  1.0000e+00+4.8986e-16j -1.0000e+00-6.1232e-16j  1.0000e+00+7.3479e-16j
  -1.0000e+00-8.5725e-16j  1.0000e+00+9.7972e-16j -1.0000e+00-1.1022e-15j  1.0000e+00+1.2246e-15j -1.0000e+00-4.8998e-15j  1.0000e+00+1.4696e-15j -1.0000e+00+1.9607e-15j
   1.0000e+00+1.7145e-15j -1.0000e+00-5.3897e-15j]
 [ 1.0000e+00+0.0000e+00j -9.2388e-01+3.8268e-01j  7.0711e-01-7.0711e-01j -3.8268e-01+9.2388e-01j  5.5109e-16-1.0000e+00j  3.8268e-01+9.2388e-01j -7.0711e-01-7.0711e-01j
   9.2388e-01+3.8268e-01j -1.0000e+00-1.1022e-15j  9.2388e-01-3.8268e-01j -7.0711e-01+7.0711e-01j  3.8268e-01-9.2388e-01j -3.4296e-15+1.0000e+00j -3.8268e-01-9.2388e-01j
   7.0711e-01+7.0711e-01j -9.2388e-01-3.8268e-01j]
 [ 1.0000e+00+0.0000e+00j -7.0711e-01+7.0711e-01j  3.0616e-16-1.0000e+00j  7.0711e-01+7.0711e-01j -1.0000e+00-6.1232e-16j  7.0711e-01-7.0711e-01j -2.6948e-15+1.0000e+00j
  -7.0711e-01-7.0711e-01j  1.0000e+00+1.2246e-15j -7.0711e-01+7.0711e-01j -2.4555e-16-1.0000e+00j  7.0711e-01+7.0711e-01j -1.0000e+00-5.3897e-15j  7.0711e-01-7.0711e-01j
  -3.9195e-15+1.0000e+00j -7.0711e-01-7.0711e-01j]
 [ 1.0000e+00+0.0000e+00j -3.8268e-01+9.2388e-01j -7.0711e-01-7.0711e-01j  9.2388e-01-3.8268e-01j -2.4499e-15+1.0000e+00j -9.2388e-01-3.8268e-01j  7.0711e-01-7.0711e-01j
   3.8268e-01+9.2388e-01j -1.0000e+00-4.8998e-15j  3.8268e-01-9.2388e-01j  7.0711e-01+7.0711e-01j -9.2388e-01+3.8268e-01j  2.4431e-16-1.0000e+00j  9.2388e-01+3.8268e-01j
  -7.0711e-01+7.0711e-01j -3.8268e-01-9.2388e-01j]
 [ 1.0000e+00+0.0000e+00j -1.8370e-16+1.0000e+00j -1.0000e+00-3.6739e-16j  5.5109e-16-1.0000e+00j  1.0000e+00+7.3479e-16j -2.6948e-15+1.0000e+00j -1.0000e+00-1.1022e-15j
  -4.9048e-16-1.0000e+00j  1.0000e+00+1.4696e-15j -3.4296e-15+1.0000e+00j -1.0000e+00-5.3897e-15j  2.4431e-16-1.0000e+00j  1.0000e+00+2.2044e-15j -4.1644e-15+1.0000e+00j
  -1.0000e+00+9.8096e-16j  9.7910e-16-1.0000e+00j]
 [ 1.0000e+00+0.0000e+00j  3.8268e-01+9.2388e-01j -7.0711e-01+7.0711e-01j -9.2388e-01-3.8268e-01j -9.8034e-16-1.0000e+00j  9.2388e-01-3.8268e-01j  7.0711e-01+7.0711e-01j
  -3.8268e-01+9.2388e-01j -1.0000e+00+1.9607e-15j -3.8268e-01-9.2388e-01j  7.0711e-01-7.0711e-01j  9.2388e-01+3.8268e-01j -4.1644e-15+1.0000e+00j -9.2388e-01+3.8268e-01j
  -7.0711e-01-7.0711e-01j  3.8268e-01-9.2388e-01j]
 [ 1.0000e+00+0.0000e+00j  7.0711e-01+7.0711e-01j -4.2863e-16+1.0000e+00j -7.0711e-01+7.0711e-01j -1.0000e+00-8.5725e-16j -7.0711e-01-7.0711e-01j -4.9048e-16-1.0000e+00j
   7.0711e-01-7.0711e-01j  1.0000e+00+1.7145e-15j  7.0711e-01+7.0711e-01j -3.9195e-15+1.0000e+00j -7.0711e-01+7.0711e-01j -1.0000e+00+9.8096e-16j -7.0711e-01-7.0711e-01j
   8.3295e-15-1.0000e+00j  7.0711e-01-7.0711e-01j]
 [ 1.0000e+00+0.0000e+00j  9.2388e-01+3.8268e-01j  7.0711e-01+7.0711e-01j  3.8268e-01+9.2388e-01j -2.6948e-15+1.0000e+00j -3.8268e-01+9.2388e-01j -7.0711e-01+7.0711e-01j
  -9.2388e-01+3.8268e-01j -1.0000e+00-5.3897e-15j -9.2388e-01-3.8268e-01j -7.0711e-01-7.0711e-01j -3.8268e-01-9.2388e-01j  9.7910e-16-1.0000e+00j  3.8268e-01-9.2388e-01j
   7.0711e-01-7.0711e-01j  9.2388e-01-3.8268e-01j]]
K =
[[ 6.2500e-02-0.0000e+00j  6.2500e-02-0.0000e+00j  6.2500e-02-0.0000e+00j  6.2500e-02-0.0000e+00j  6.2500e-02-0.0000e+00j  6.2500e-02-0.0000e+00j  6.2500e-02-0.0000e+00j
   6.2500e-02-0.0000e+00j  6.2500e-02-0.0000e+00j  6.2500e-02-0.0000e+00j  6.2500e-02-0.0000e+00j  6.2500e-02-0.0000e+00j  6.2500e-02-0.0000e+00j  6.2500e-02-0.0000e+00j
   6.2500e-02-0.0000e+00j  6.2500e-02-0.0000e+00j]
 [ 6.2500e-02-0.0000e+00j  5.7742e-02+2.3918e-02j  4.4194e-02+4.4194e-02j  2.3918e-02+5.7742e-02j  3.8270e-18+6.2500e-02j -2.3918e-02+5.7742e-02j -4.4194e-02+4.4194e-02j
  -5.7742e-02+2.3918e-02j -6.2500e-02+7.6540e-18j -5.7742e-02-2.3918e-02j -4.4194e-02-4.4194e-02j -2.3918e-02-5.7742e-02j -1.1481e-17-6.2500e-02j  2.3918e-02-5.7742e-02j
   4.4194e-02-4.4194e-02j  5.7742e-02-2.3918e-02j]
 [ 6.2500e-02-0.0000e+00j  4.4194e-02+4.4194e-02j  3.8270e-18+6.2500e-02j -4.4194e-02+4.4194e-02j -6.2500e-02+7.6540e-18j -4.4194e-02-4.4194e-02j -1.1481e-17-6.2500e-02j
   4.4194e-02-4.4194e-02j  6.2500e-02-1.5308e-17j  4.4194e-02+4.4194e-02j  1.9135e-17+6.2500e-02j -4.4194e-02+4.4194e-02j -6.2500e-02+2.2962e-17j -4.4194e-02-4.4194e-02j
  -2.6789e-17-6.2500e-02j  4.4194e-02-4.4194e-02j]
 [ 6.2500e-02-0.0000e+00j  2.3918e-02+5.7742e-02j -4.4194e-02+4.4194e-02j -5.7742e-02-2.3918e-02j -1.1481e-17-6.2500e-02j  5.7742e-02-2.3918e-02j  4.4194e-02+4.4194e-02j
  -2.3918e-02+5.7742e-02j -6.2500e-02+2.2962e-17j -2.3918e-02-5.7742e-02j  4.4194e-02-4.4194e-02j  5.7742e-02+2.3918e-02j  3.4443e-17+6.2500e-02j -5.7742e-02+2.3918e-02j
  -4.4194e-02-4.4194e-02j  2.3918e-02-5.7742e-02j]
 [ 6.2500e-02-0.0000e+00j  3.8270e-18+6.2500e-02j -6.2500e-02+7.6540e-18j -1.1481e-17-6.2500e-02j  6.2500e-02-1.5308e-17j  1.9135e-17+6.2500e-02j -6.2500e-02+2.2962e-17j
  -2.6789e-17-6.2500e-02j  6.2500e-02-3.0616e-17j  3.4443e-17+6.2500e-02j -6.2500e-02+3.8270e-17j -1.5312e-16-6.2500e-02j  6.2500e-02-4.5924e-17j -6.1271e-17+6.2500e-02j
  -6.2500e-02+5.3578e-17j -1.6843e-16-6.2500e-02j]
 [ 6.2500e-02-0.0000e+00j -2.3918e-02+5.7742e-02j -4.4194e-02-4.4194e-02j  5.7742e-02-2.3918e-02j  1.9135e-17+6.2500e-02j -5.7742e-02-2.3918e-02j  4.4194e-02-4.4194e-02j
   2.3918e-02+5.7742e-02j -6.2500e-02+3.8270e-17j  2.3918e-02-5.7742e-02j  4.4194e-02+4.4194e-02j -5.7742e-02+2.3918e-02j -1.6843e-16-6.2500e-02j  5.7742e-02+2.3918e-02j
  -4.4194e-02+4.4194e-02j -2.3918e-02-5.7742e-02j]
 [ 6.2500e-02-0.0000e+00j -4.4194e-02+4.4194e-02j -1.1481e-17-6.2500e-02j  4.4194e-02+4.4194e-02j -6.2500e-02+2.2962e-17j  4.4194e-02-4.4194e-02j  3.4443e-17+6.2500e-02j
  -4.4194e-02-4.4194e-02j  6.2500e-02-4.5924e-17j -4.4194e-02+4.4194e-02j -1.6843e-16-6.2500e-02j  4.4194e-02+4.4194e-02j -6.2500e-02+6.8886e-17j  4.4194e-02-4.4194e-02j
  -3.0655e-17+6.2500e-02j -4.4194e-02-4.4194e-02j]
 [ 6.2500e-02-0.0000e+00j -5.7742e-02+2.3918e-02j  4.4194e-02-4.4194e-02j -2.3918e-02+5.7742e-02j -2.6789e-17-6.2500e-02j  2.3918e-02+5.7742e-02j -4.4194e-02-4.4194e-02j
   5.7742e-02+2.3918e-02j -6.2500e-02+5.3578e-17j  5.7742e-02-2.3918e-02j -4.4194e-02+4.4194e-02j  2.3918e-02-5.7742e-02j -3.0655e-17+6.2500e-02j -2.3918e-02-5.7742e-02j
   4.4194e-02+4.4194e-02j -5.7742e-02-2.3918e-02j]
 [ 6.2500e-02-0.0000e+00j -6.2500e-02+7.6540e-18j  6.2500e-02-1.5308e-17j -6.2500e-02+2.2962e-17j  6.2500e-02-3.0616e-17j -6.2500e-02+3.8270e-17j  6.2500e-02-4.5924e-17j
  -6.2500e-02+5.3578e-17j  6.2500e-02-6.1232e-17j -6.2500e-02+6.8886e-17j  6.2500e-02-7.6540e-17j -6.2500e-02+3.0624e-16j  6.2500e-02-9.1849e-17j -6.2500e-02-1.2254e-16j
   6.2500e-02-1.0716e-16j -6.2500e-02+3.3686e-16j]
 [ 6.2500e-02-0.0000e+00j -5.7742e-02-2.3918e-02j  4.4194e-02+4.4194e-02j -2.3918e-02-5.7742e-02j  3.4443e-17+6.2500e-02j  2.3918e-02-5.7742e-02j -4.4194e-02+4.4194e-02j
   5.7742e-02-2.3918e-02j -6.2500e-02+6.8886e-17j  5.7742e-02+2.3918e-02j -4.4194e-02-4.4194e-02j  2.3918e-02+5.7742e-02j -2.1435e-16-6.2500e-02j -2.3918e-02+5.7742e-02j
   4.4194e-02-4.4194e-02j -5.7742e-02+2.3918e-02j]
 [ 6.2500e-02-0.0000e+00j -4.4194e-02-4.4194e-02j  1.9135e-17+6.2500e-02j  4.4194e-02-4.4194e-02j -6.2500e-02+3.8270e-17j  4.4194e-02+4.4194e-02j -1.6843e-16-6.2500e-02j
  -4.4194e-02+4.4194e-02j  6.2500e-02-7.6540e-17j -4.4194e-02-4.4194e-02j -1.5347e-17+6.2500e-02j  4.4194e-02-4.4194e-02j -6.2500e-02+3.3686e-16j  4.4194e-02+4.4194e-02j
  -2.4497e-16-6.2500e-02j -4.4194e-02+4.4194e-02j]
 [ 6.2500e-02-0.0000e+00j -2.3918e-02-5.7742e-02j -4.4194e-02+4.4194e-02j  5.7742e-02+2.3918e-02j -1.5312e-16-6.2500e-02j -5.7742e-02+2.3918e-02j  4.4194e-02+4.4194e-02j
   2.3918e-02-5.7742e-02j -6.2500e-02+3.0624e-16j  2.3918e-02+5.7742e-02j  4.4194e-02-4.4194e-02j -5.7742e-02-2.3918e-02j  1.5269e-17+6.2500e-02j  5.7742e-02-2.3918e-02j
  -4.4194e-02-4.4194e-02j -2.3918e-02+5.7742e-02j]
 [ 6.2500e-02-0.0000e+00j -1.1481e-17-6.2500e-02j -6.2500e-02+2.2962e-17j  3.4443e-17+6.2500e-02j  6.2500e-02-4.5924e-17j -1.6843e-16-6.2500e-02j -6.2500e-02+6.8886e-17j
  -3.0655e-17+6.2500e-02j  6.2500e-02-9.1849e-17j -2.1435e-16-6.2500e-02j -6.2500e-02+3.3686e-16j  1.5269e-17+6.2500e-02j  6.2500e-02-1.3777e-16j -2.6028e-16-6.2500e-02j
  -6.2500e-02-6.1310e-17j  6.1194e-17+6.2500e-02j]
 [ 6.2500e-02-0.0000e+00j  2.3918e-02-5.7742e-02j -4.4194e-02-4.4194e-02j -5.7742e-02+2.3918e-02j -6.1271e-17+6.2500e-02j  5.7742e-02+2.3918e-02j  4.4194e-02-4.4194e-02j
  -2.3918e-02-5.7742e-02j -6.2500e-02-1.2254e-16j -2.3918e-02+5.7742e-02j  4.4194e-02+4.4194e-02j  5.7742e-02-2.3918e-02j -2.6028e-16-6.2500e-02j -5.7742e-02-2.3918e-02j
  -4.4194e-02+4.4194e-02j  2.3918e-02+5.7742e-02j]
 [ 6.2500e-02-0.0000e+00j  4.4194e-02-4.4194e-02j -2.6789e-17-6.2500e-02j -4.4194e-02-4.4194e-02j -6.2500e-02+5.3578e-17j -4.4194e-02+4.4194e-02j -3.0655e-17+6.2500e-02j
   4.4194e-02+4.4194e-02j  6.2500e-02-1.0716e-16j  4.4194e-02-4.4194e-02j -2.4497e-16-6.2500e-02j -4.4194e-02-4.4194e-02j -6.2500e-02-6.1310e-17j -4.4194e-02+4.4194e-02j
   5.2059e-16+6.2500e-02j  4.4194e-02+4.4194e-02j]
 [ 6.2500e-02-0.0000e+00j  5.7742e-02-2.3918e-02j  4.4194e-02-4.4194e-02j  2.3918e-02-5.7742e-02j -1.6843e-16-6.2500e-02j -2.3918e-02-5.7742e-02j -4.4194e-02-4.4194e-02j
  -5.7742e-02-2.3918e-02j -6.2500e-02+3.3686e-16j -5.7742e-02+2.3918e-02j -4.4194e-02+4.4194e-02j -2.3918e-02+5.7742e-02j  6.1194e-17+6.2500e-02j  2.3918e-02+5.7742e-02j
   4.4194e-02+4.4194e-02j  5.7742e-02+2.3918e-02j]]
x =
[ 1.4375+0.0000e+00j  0.6433+8.7881e-01j -0.096 +2.9419e-01j  0.3038-1.2516e-01j  0.375 +1.8750e-01j  0.0926+2.1284e-02j  0.346 -2.0581e-01j  0.4602+2.5256e-02j
  0.3125+5.3578e-17j  0.4602-2.5256e-02j  0.346 +2.0581e-01j  0.0926-2.1284e-02j  0.375 -1.8750e-01j  0.3038+1.2516e-01j -0.096 -2.9419e-01j  0.6433-8.7881e-01j]

N = 20
W =
[[ 1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j
   1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j
   1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j  1.0000e+00+0.0000e+00j]
 [ 1.0000e+00+0.0000e+00j  9.5106e-01-3.0902e-01j  8.0902e-01-5.8779e-01j  5.8779e-01-8.0902e-01j  3.0902e-01-9.5106e-01j  6.1232e-17-1.0000e+00j -3.0902e-01-9.5106e-01j
  -5.8779e-01-8.0902e-01j -8.0902e-01-5.8779e-01j -9.5106e-01-3.0902e-01j -1.0000e+00-1.2246e-16j -9.5106e-01+3.0902e-01j -8.0902e-01+5.8779e-01j -5.8779e-01+8.0902e-01j
  -3.0902e-01+9.5106e-01j -1.8370e-16+1.0000e+00j  3.0902e-01+9.5106e-01j  5.8779e-01+8.0902e-01j  8.0902e-01+5.8779e-01j  9.5106e-01+3.0902e-01j]
 [ 1.0000e+00+0.0000e+00j  8.0902e-01-5.8779e-01j  3.0902e-01-9.5106e-01j -3.0902e-01-9.5106e-01j -8.0902e-01-5.8779e-01j -1.0000e+00-1.2246e-16j -8.0902e-01+5.8779e-01j
  -3.0902e-01+9.5106e-01j  3.0902e-01+9.5106e-01j  8.0902e-01+5.8779e-01j  1.0000e+00+2.4493e-16j  8.0902e-01-5.8779e-01j  3.0902e-01-9.5106e-01j -3.0902e-01-9.5106e-01j
  -8.0902e-01-5.8779e-01j -1.0000e+00-3.6739e-16j -8.0902e-01+5.8779e-01j -3.0902e-01+9.5106e-01j  3.0902e-01+9.5106e-01j  8.0902e-01+5.8779e-01j]
 [ 1.0000e+00+0.0000e+00j  5.8779e-01-8.0902e-01j -3.0902e-01-9.5106e-01j -9.5106e-01-3.0902e-01j -8.0902e-01+5.8779e-01j -1.8370e-16+1.0000e+00j  8.0902e-01+5.8779e-01j
   9.5106e-01-3.0902e-01j  3.0902e-01-9.5106e-01j -5.8779e-01-8.0902e-01j -1.0000e+00-3.6739e-16j -5.8779e-01+8.0902e-01j  3.0902e-01+9.5106e-01j  9.5106e-01+3.0902e-01j
   8.0902e-01-5.8779e-01j -1.2253e-15-1.0000e+00j -8.0902e-01-5.8779e-01j -9.5106e-01+3.0902e-01j -3.0902e-01+9.5106e-01j  5.8779e-01+8.0902e-01j]
 [ 1.0000e+00+0.0000e+00j  3.0902e-01-9.5106e-01j -8.0902e-01-5.8779e-01j -8.0902e-01+5.8779e-01j  3.0902e-01+9.5106e-01j  1.0000e+00+2.4493e-16j  3.0902e-01-9.5106e-01j
  -8.0902e-01-5.8779e-01j -8.0902e-01+5.8779e-01j  3.0902e-01+9.5106e-01j  1.0000e+00+4.8986e-16j  3.0902e-01-9.5106e-01j -8.0902e-01-5.8779e-01j -8.0902e-01+5.8779e-01j
   3.0902e-01+9.5106e-01j  1.0000e+00+7.3479e-16j  3.0902e-01-9.5106e-01j -8.0902e-01-5.8779e-01j -8.0902e-01+5.8779e-01j  3.0902e-01+9.5106e-01j]
 [ 1.0000e+00+0.0000e+00j  6.1232e-17-1.0000e+00j -1.0000e+00-1.2246e-16j -1.8370e-16+1.0000e+00j  1.0000e+00+2.4493e-16j -5.8202e-16-1.0000e+00j -1.0000e+00-3.6739e-16j
  -4.2863e-16+1.0000e+00j  1.0000e+00+4.8986e-16j -1.2253e-15-1.0000e+00j -1.0000e+00+1.1640e-15j  1.1028e-15+1.0000e+00j  1.0000e+00+7.3479e-16j -9.8034e-16-1.0000e+00j
  -1.0000e+00-8.5725e-16j  8.5787e-16+1.0000e+00j  1.0000e+00+9.7972e-16j -7.3541e-16-1.0000e+00j -1.0000e+00+2.4505e-15j  6.1294e-16+1.0000e+00j]
 [ 1.0000e+00+0.0000e+00j -3.0902e-01-9.5106e-01j -8.0902e-01+5.8779e-01j  8.0902e-01+5.8779e-01j  3.0902e-01-9.5106e-01j -1.0000e+00-3.6739e-16j  3.0902e-01+9.5106e-01j
   8.0902e-01-5.8779e-01j -8.0902e-01-5.8779e-01j -3.0902e-01+9.5106e-01j  1.0000e+00+7.3479e-16j -3.0902e-01-9.5106e-01j -8.0902e-01+5.8779e-01j  8.0902e-01+5.8779e-01j
   3.0902e-01-9.5106e-01j -1.0000e+00+2.4505e-15j  3.0902e-01+9.5106e-01j  8.0902e-01-5.8779e-01j -8.0902e-01-5.8779e-01j -3.0902e-01+9.5106e-01j]
 [ 1.0000e+00+0.0000e+00j -5.8779e-01-8.0902e-01j -3.0902e-01+9.5106e-01j  9.5106e-01-3.0902e-01j -8.0902e-01-5.8779e-01j -4.2863e-16+1.0000e+00j  8.0902e-01-5.8779e-01j
  -9.5106e-01-3.0902e-01j  3.0902e-01+9.5106e-01j  5.8779e-01-8.0902e-01j -1.0000e+00-8.5725e-16j  5.8779e-01+8.0902e-01j  3.0902e-01-9.5106e-01j -9.5106e-01+3.0902e-01j
   8.0902e-01+5.8779e-01j -4.9048e-16-1.0000e+00j -8.0902e-01+5.8779e-01j  9.5106e-01+3.0902e-01j -3.0902e-01-9.5106e-01j -5.8779e-01+8.0902e-01j]
 [ 1.0000e+00+0.0000e+00j -8.0902e-01-5.8779e-01j  3.0902e-01+9.5106e-01j  3.0902e-01-9.5106e-01j -8.0902e-01+5.8779e-01j  1.0000e+00+4.8986e-16j -8.0902e-01-5.8779e-01j
   3.0902e-01+9.5106e-01j  3.0902e-01-9.5106e-01j -8.0902e-01+5.8779e-01j  1.0000e+00+9.7972e-16j -8.0902e-01-5.8779e-01j  3.0902e-01+9.5106e-01j  3.0902e-01-9.5106e-01j
  -8.0902e-01+5.8779e-01j  1.0000e+00+1.4696e-15j -8.0902e-01-5.8779e-01j  3.0902e-01+9.5106e-01j  3.0902e-01-9.5106e-01j -8.0902e-01+5.8779e-01j]
 [ 1.0000e+00+0.0000e+00j -9.5106e-01-3.0902e-01j  8.0902e-01+5.8779e-01j -5.8779e-01-8.0902e-01j  3.0902e-01+9.5106e-01j -1.2253e-15-1.0000e+00j -3.0902e-01+9.5106e-01j
   5.8779e-01-8.0902e-01j -8.0902e-01+5.8779e-01j  9.5106e-01-3.0902e-01j -1.0000e+00+2.4505e-15j  9.5106e-01+3.0902e-01j -8.0902e-01-5.8779e-01j  5.8779e-01+8.0902e-01j
  -3.0902e-01-9.5106e-01j -3.4296e-15+1.0000e+00j  3.0902e-01-9.5106e-01j -5.8779e-01+8.0902e-01j  8.0902e-01-5.8779e-01j -9.5106e-01+3.0902e-01j]
 [ 1.0000e+00+0.0000e+00j -1.0000e+00-1.2246e-16j  1.0000e+00+2.4493e-16j -1.0000e+00-3.6739e-16j  1.0000e+00+4.8986e-16j -1.0000e+00+1.1640e-15j  1.0000e+00+7.3479e-16j
  -1.0000e+00-8.5725e-16j  1.0000e+00+9.7972e-16j -1.0000e+00+2.4505e-15j  1.0000e+00-2.3281e-15j -1.0000e+00+2.2056e-15j  1.0000e+00+1.4696e-15j -1.0000e+00+1.9607e-15j
   1.0000e+00+1.7145e-15j -1.0000e+00+1.7157e-15j  1.0000e+00+1.9594e-15j -1.0000e+00+1.4708e-15j  1.0000e+00-4.9011e-15j -1.0000e+00+1.2259e-15j]
 [ 1.0000e+00+0.0000e+00j -9.5106e-01+3.0902e-01j  8.0902e-01-5.8779e-01j -5.8779e-01+8.0902e-01j  3.0902e-01-9.5106e-01j  1.1028e-15+1.0000e+00j -3.0902e-01-9.5106e-01j
   5.8779e-01+8.0902e-01j -8.0902e-01-5.8779e-01j  9.5106e-01+3.0902e-01j -1.0000e+00+2.2056e-15j  9.5106e-01-3.0902e-01j -8.0902e-01+5.8779e-01j  5.8779e-01-8.0902e-01j
  -3.0902e-01+9.5106e-01j  7.3497e-15-1.0000e+00j  3.0902e-01+9.5106e-01j -5.8779e-01-8.0902e-01j  8.0902e-01+5.8779e-01j -9.5106e-01-3.0902e-01j]
 [ 1.0000e+00+0.0000e+00j -8.0902e-01+5.8779e-01j  3.0902e-01-9.5106e-01j  3.0902e-01+9.5106e-01j -8.0902e-01-5.8779e-01j  1.0000e+00+7.3479e-16j -8.0902e-01+5.8779e-01j
   3.0902e-01-9.5106e-01j  3.0902e-01+9.5106e-01j -8.0902e-01-5.8779e-01j  1.0000e+00+1.4696e-15j -8.0902e-01+5.8779e-01j  3.0902e-01-9.5106e-01j  3.0902e-01+9.5106e-01j
  -8.0902e-01-5.8779e-01j  1.0000e+00-4.9011e-15j -8.0902e-01+5.8779e-01j  3.0902e-01-9.5106e-01j  3.0902e-01+9.5106e-01j -8.0902e-01-5.8779e-01j]
 [ 1.0000e+00+0.0000e+00j -5.8779e-01+8.0902e-01j -3.0902e-01-9.5106e-01j  9.5106e-01+3.0902e-01j -8.0902e-01+5.8779e-01j -9.8034e-16-1.0000e+00j  8.0902e-01+5.8779e-01j
  -9.5106e-01+3.0902e-01j  3.0902e-01-9.5106e-01j  5.8779e-01+8.0902e-01j -1.0000e+00+1.9607e-15j  5.8779e-01-8.0902e-01j  3.0902e-01+9.5106e-01j -9.5106e-01-3.0902e-01j
   8.0902e-01-5.8779e-01j -4.1644e-15+1.0000e+00j -8.0902e-01-5.8779e-01j  9.5106e-01-3.0902e-01j -3.0902e-01+9.5106e-01j -5.8779e-01-8.0902e-01j]
 [ 1.0000e+00+0.0000e+00j -3.0902e-01+9.5106e-01j -8.0902e-01-5.8779e-01j  8.0902e-01-5.8779e-01j  3.0902e-01+9.5106e-01j -1.0000e+00-8.5725e-16j  3.0902e-01-9.5106e-01j
   8.0902e-01+5.8779e-01j -8.0902e-01+5.8779e-01j -3.0902e-01-9.5106e-01j  1.0000e+00+1.7145e-15j -3.0902e-01+9.5106e-01j -8.0902e-01-5.8779e-01j  8.0902e-01-5.8779e-01j
   3.0902e-01+9.5106e-01j -1.0000e+00+9.8096e-16j  3.0902e-01-9.5106e-01j  8.0902e-01+5.8779e-01j -8.0902e-01+5.8779e-01j -3.0902e-01-9.5106e-01j]
 [ 1.0000e+00+0.0000e+00j -1.8370e-16+1.0000e+00j -1.0000e+00-3.6739e-16j -1.2253e-15-1.0000e+00j  1.0000e+00+7.3479e-16j  8.5787e-16+1.0000e+00j -1.0000e+00+2.4505e-15j
  -4.9048e-16-1.0000e+00j  1.0000e+00+1.4696e-15j -3.4296e-15+1.0000e+00j -1.0000e+00+1.7157e-15j  7.3497e-15-1.0000e+00j  1.0000e+00-4.9011e-15j -4.1644e-15+1.0000e+00j
  -1.0000e+00+9.8096e-16j  9.7910e-16-1.0000e+00j  1.0000e+00+2.9392e-15j  9.3116e-15+1.0000e+00j -1.0000e+00-6.8593e-15j -5.3915e-15-1.0000e+00j]
 [ 1.0000e+00+0.0000e+00j  3.0902e-01+9.5106e-01j -8.0902e-01+5.8779e-01j -8.0902e-01-5.8779e-01j  3.0902e-01-9.5106e-01j  1.0000e+00+9.7972e-16j  3.0902e-01+9.5106e-01j
  -8.0902e-01+5.8779e-01j -8.0902e-01-5.8779e-01j  3.0902e-01-9.5106e-01j  1.0000e+00+1.9594e-15j  3.0902e-01+9.5106e-01j -8.0902e-01+5.8779e-01j -8.0902e-01-5.8779e-01j
   3.0902e-01-9.5106e-01j  1.0000e+00+2.9392e-15j  3.0902e-01+9.5106e-01j -8.0902e-01+5.8779e-01j -8.0902e-01-5.8779e-01j  3.0902e-01-9.5106e-01j]
 [ 1.0000e+00+0.0000e+00j  5.8779e-01+8.0902e-01j -3.0902e-01+9.5106e-01j -9.5106e-01+3.0902e-01j -8.0902e-01-5.8779e-01j -7.3541e-16-1.0000e+00j  8.0902e-01-5.8779e-01j
   9.5106e-01+3.0902e-01j  3.0902e-01+9.5106e-01j -5.8779e-01+8.0902e-01j -1.0000e+00+1.4708e-15j -5.8779e-01-8.0902e-01j  3.0902e-01-9.5106e-01j  9.5106e-01-3.0902e-01j
   8.0902e-01+5.8779e-01j  9.3116e-15+1.0000e+00j -8.0902e-01+5.8779e-01j -9.5106e-01-3.0902e-01j -3.0902e-01-9.5106e-01j  5.8779e-01-8.0902e-01j]
 [ 1.0000e+00+0.0000e+00j  8.0902e-01+5.8779e-01j  3.0902e-01+9.5106e-01j -3.0902e-01+9.5106e-01j -8.0902e-01+5.8779e-01j -1.0000e+00+2.4505e-15j -8.0902e-01-5.8779e-01j
  -3.0902e-01-9.5106e-01j  3.0902e-01-9.5106e-01j  8.0902e-01-5.8779e-01j  1.0000e+00-4.9011e-15j  8.0902e-01+5.8779e-01j  3.0902e-01+9.5106e-01j -3.0902e-01+9.5106e-01j
  -8.0902e-01+5.8779e-01j -1.0000e+00-6.8593e-15j -8.0902e-01-5.8779e-01j -3.0902e-01-9.5106e-01j  3.0902e-01-9.5106e-01j  8.0902e-01-5.8779e-01j]
 [ 1.0000e+00+0.0000e+00j  9.5106e-01+3.0902e-01j  8.0902e-01+5.8779e-01j  5.8779e-01+8.0902e-01j  3.0902e-01+9.5106e-01j  6.1294e-16+1.0000e+00j -3.0902e-01+9.5106e-01j
  -5.8779e-01+8.0902e-01j -8.0902e-01+5.8779e-01j -9.5106e-01+3.0902e-01j -1.0000e+00+1.2259e-15j -9.5106e-01-3.0902e-01j -8.0902e-01-5.8779e-01j -5.8779e-01-8.0902e-01j
  -3.0902e-01-9.5106e-01j -5.3915e-15-1.0000e+00j  3.0902e-01-9.5106e-01j  5.8779e-01-8.0902e-01j  8.0902e-01-5.8779e-01j  9.5106e-01-3.0902e-01j]]
K =
[[ 5.0000e-02-0.0000e+00j  5.0000e-02-0.0000e+00j  5.0000e-02-0.0000e+00j  5.0000e-02-0.0000e+00j  5.0000e-02-0.0000e+00j  5.0000e-02-0.0000e+00j  5.0000e-02-0.0000e+00j
   5.0000e-02-0.0000e+00j  5.0000e-02-0.0000e+00j  5.0000e-02-0.0000e+00j  5.0000e-02-0.0000e+00j  5.0000e-02-0.0000e+00j  5.0000e-02-0.0000e+00j  5.0000e-02-0.0000e+00j
   5.0000e-02-0.0000e+00j  5.0000e-02-0.0000e+00j  5.0000e-02-0.0000e+00j  5.0000e-02-0.0000e+00j  5.0000e-02-0.0000e+00j  5.0000e-02-0.0000e+00j]
 [ 5.0000e-02-0.0000e+00j  4.7553e-02+1.5451e-02j  4.0451e-02+2.9389e-02j  2.9389e-02+4.0451e-02j  1.5451e-02+4.7553e-02j  3.0616e-18+5.0000e-02j -1.5451e-02+4.7553e-02j
  -2.9389e-02+4.0451e-02j -4.0451e-02+2.9389e-02j -4.7553e-02+1.5451e-02j -5.0000e-02+6.1232e-18j -4.7553e-02-1.5451e-02j -4.0451e-02-2.9389e-02j -2.9389e-02-4.0451e-02j
  -1.5451e-02-4.7553e-02j -9.1849e-18-5.0000e-02j  1.5451e-02-4.7553e-02j  2.9389e-02-4.0451e-02j  4.0451e-02-2.9389e-02j  4.7553e-02-1.5451e-02j]
 [ 5.0000e-02-0.0000e+00j  4.0451e-02+2.9389e-02j  1.5451e-02+4.7553e-02j -1.5451e-02+4.7553e-02j -4.0451e-02+2.9389e-02j -5.0000e-02+6.1232e-18j -4.0451e-02-2.9389e-02j
  -1.5451e-02-4.7553e-02j  1.5451e-02-4.7553e-02j  4.0451e-02-2.9389e-02j  5.0000e-02-1.2246e-17j  4.0451e-02+2.9389e-02j  1.5451e-02+4.7553e-02j -1.5451e-02+4.7553e-02j
  -4.0451e-02+2.9389e-02j -5.0000e-02+1.8370e-17j -4.0451e-02-2.9389e-02j -1.5451e-02-4.7553e-02j  1.5451e-02-4.7553e-02j  4.0451e-02-2.9389e-02j]
 [ 5.0000e-02-0.0000e+00j  2.9389e-02+4.0451e-02j -1.5451e-02+4.7553e-02j -4.7553e-02+1.5451e-02j -4.0451e-02-2.9389e-02j -9.1849e-18-5.0000e-02j  4.0451e-02-2.9389e-02j
   4.7553e-02+1.5451e-02j  1.5451e-02+4.7553e-02j -2.9389e-02+4.0451e-02j -5.0000e-02+1.8370e-17j -2.9389e-02-4.0451e-02j  1.5451e-02-4.7553e-02j  4.7553e-02-1.5451e-02j
   4.0451e-02+2.9389e-02j -6.1263e-17+5.0000e-02j -4.0451e-02+2.9389e-02j -4.7553e-02-1.5451e-02j -1.5451e-02-4.7553e-02j  2.9389e-02-4.0451e-02j]
 [ 5.0000e-02-0.0000e+00j  1.5451e-02+4.7553e-02j -4.0451e-02+2.9389e-02j -4.0451e-02-2.9389e-02j  1.5451e-02-4.7553e-02j  5.0000e-02-1.2246e-17j  1.5451e-02+4.7553e-02j
  -4.0451e-02+2.9389e-02j -4.0451e-02-2.9389e-02j  1.5451e-02-4.7553e-02j  5.0000e-02-2.4493e-17j  1.5451e-02+4.7553e-02j -4.0451e-02+2.9389e-02j -4.0451e-02-2.9389e-02j
   1.5451e-02-4.7553e-02j  5.0000e-02-3.6739e-17j  1.5451e-02+4.7553e-02j -4.0451e-02+2.9389e-02j -4.0451e-02-2.9389e-02j  1.5451e-02-4.7553e-02j]
 [ 5.0000e-02-0.0000e+00j  3.0616e-18+5.0000e-02j -5.0000e-02+6.1232e-18j -9.1849e-18-5.0000e-02j  5.0000e-02-1.2246e-17j -2.9101e-17+5.0000e-02j -5.0000e-02+1.8370e-17j
  -2.1431e-17-5.0000e-02j  5.0000e-02-2.4493e-17j -6.1263e-17+5.0000e-02j -5.0000e-02-5.8202e-17j  5.5140e-17-5.0000e-02j  5.0000e-02-3.6739e-17j -4.9017e-17+5.0000e-02j
  -5.0000e-02+4.2863e-17j  4.2894e-17-5.0000e-02j  5.0000e-02-4.8986e-17j -3.6770e-17+5.0000e-02j -5.0000e-02-1.2253e-16j  3.0647e-17-5.0000e-02j]
 [ 5.0000e-02-0.0000e+00j -1.5451e-02+4.7553e-02j -4.0451e-02-2.9389e-02j  4.0451e-02-2.9389e-02j  1.5451e-02+4.7553e-02j -5.0000e-02+1.8370e-17j  1.5451e-02-4.7553e-02j
   4.0451e-02+2.9389e-02j -4.0451e-02+2.9389e-02j -1.5451e-02-4.7553e-02j  5.0000e-02-3.6739e-17j -1.5451e-02+4.7553e-02j -4.0451e-02-2.9389e-02j  4.0451e-02-2.9389e-02j
   1.5451e-02+4.7553e-02j -5.0000e-02-1.2253e-16j  1.5451e-02-4.7553e-02j  4.0451e-02+2.9389e-02j -4.0451e-02+2.9389e-02j -1.5451e-02-4.7553e-02j]
 [ 5.0000e-02-0.0000e+00j -2.9389e-02+4.0451e-02j -1.5451e-02-4.7553e-02j  4.7553e-02+1.5451e-02j -4.0451e-02+2.9389e-02j -2.1431e-17-5.0000e-02j  4.0451e-02+2.9389e-02j
  -4.7553e-02+1.5451e-02j  1.5451e-02-4.7553e-02j  2.9389e-02+4.0451e-02j -5.0000e-02+4.2863e-17j  2.9389e-02-4.0451e-02j  1.5451e-02+4.7553e-02j -4.7553e-02-1.5451e-02j
   4.0451e-02-2.9389e-02j -2.4524e-17+5.0000e-02j -4.0451e-02-2.9389e-02j  4.7553e-02-1.5451e-02j -1.5451e-02+4.7553e-02j -2.9389e-02-4.0451e-02j]
 [ 5.0000e-02-0.0000e+00j -4.0451e-02+2.9389e-02j  1.5451e-02-4.7553e-02j  1.5451e-02+4.7553e-02j -4.0451e-02-2.9389e-02j  5.0000e-02-2.4493e-17j -4.0451e-02+2.9389e-02j
   1.5451e-02-4.7553e-02j  1.5451e-02+4.7553e-02j -4.0451e-02-2.9389e-02j  5.0000e-02-4.8986e-17j -4.0451e-02+2.9389e-02j  1.5451e-02-4.7553e-02j  1.5451e-02+4.7553e-02j
  -4.0451e-02-2.9389e-02j  5.0000e-02-7.3479e-17j -4.0451e-02+2.9389e-02j  1.5451e-02-4.7553e-02j  1.5451e-02+4.7553e-02j -4.0451e-02-2.9389e-02j]
 [ 5.0000e-02-0.0000e+00j -4.7553e-02+1.5451e-02j  4.0451e-02-2.9389e-02j -2.9389e-02+4.0451e-02j  1.5451e-02-4.7553e-02j -6.1263e-17+5.0000e-02j -1.5451e-02-4.7553e-02j
   2.9389e-02+4.0451e-02j -4.0451e-02-2.9389e-02j  4.7553e-02+1.5451e-02j -5.0000e-02-1.2253e-16j  4.7553e-02-1.5451e-02j -4.0451e-02+2.9389e-02j  2.9389e-02-4.0451e-02j
  -1.5451e-02+4.7553e-02j -1.7148e-16-5.0000e-02j  1.5451e-02+4.7553e-02j -2.9389e-02-4.0451e-02j  4.0451e-02+2.9389e-02j -4.7553e-02-1.5451e-02j]
 [ 5.0000e-02-0.0000e+00j -5.0000e-02+6.1232e-18j  5.0000e-02-1.2246e-17j -5.0000e-02+1.8370e-17j  5.0000e-02-2.4493e-17j -5.0000e-02-5.8202e-17j  5.0000e-02-3.6739e-17j
  -5.0000e-02+4.2863e-17j  5.0000e-02-4.8986e-17j -5.0000e-02-1.2253e-16j  5.0000e-02+1.1640e-16j -5.0000e-02-1.1028e-16j  5.0000e-02-7.3479e-17j -5.0000e-02-9.8034e-17j
   5.0000e-02-8.5725e-17j -5.0000e-02-8.5787e-17j  5.0000e-02-9.7972e-17j -5.0000e-02-7.3541e-17j  5.0000e-02+2.4505e-16j -5.0000e-02-6.1294e-17j]
 [ 5.0000e-02-0.0000e+00j -4.7553e-02-1.5451e-02j  4.0451e-02+2.9389e-02j -2.9389e-02-4.0451e-02j  1.5451e-02+4.7553e-02j  5.5140e-17-5.0000e-02j -1.5451e-02+4.7553e-02j
   2.9389e-02-4.0451e-02j -4.0451e-02+2.9389e-02j  4.7553e-02-1.5451e-02j -5.0000e-02-1.1028e-16j  4.7553e-02+1.5451e-02j -4.0451e-02-2.9389e-02j  2.9389e-02+4.0451e-02j
  -1.5451e-02-4.7553e-02j  3.6749e-16+5.0000e-02j  1.5451e-02-4.7553e-02j -2.9389e-02+4.0451e-02j  4.0451e-02-2.9389e-02j -4.7553e-02+1.5451e-02j]
 [ 5.0000e-02-0.0000e+00j -4.0451e-02-2.9389e-02j  1.5451e-02+4.7553e-02j  1.5451e-02-4.7553e-02j -4.0451e-02+2.9389e-02j  5.0000e-02-3.6739e-17j -4.0451e-02-2.9389e-02j
   1.5451e-02+4.7553e-02j  1.5451e-02-4.7553e-02j -4.0451e-02+2.9389e-02j  5.0000e-02-7.3479e-17j -4.0451e-02-2.9389e-02j  1.5451e-02+4.7553e-02j  1.5451e-02-4.7553e-02j
  -4.0451e-02+2.9389e-02j  5.0000e-02+2.4505e-16j -4.0451e-02-2.9389e-02j  1.5451e-02+4.7553e-02j  1.5451e-02-4.7553e-02j -4.0451e-02+2.9389e-02j]
 [ 5.0000e-02-0.0000e+00j -2.9389e-02-4.0451e-02j -1.5451e-02+4.7553e-02j  4.7553e-02-1.5451e-02j -4.0451e-02-2.9389e-02j -4.9017e-17+5.0000e-02j  4.0451e-02-2.9389e-02j
  -4.7553e-02-1.5451e-02j  1.5451e-02+4.7553e-02j  2.9389e-02-4.0451e-02j -5.0000e-02-9.8034e-17j  2.9389e-02+4.0451e-02j  1.5451e-02-4.7553e-02j -4.7553e-02+1.5451e-02j
   4.0451e-02+2.9389e-02j -2.0822e-16-5.0000e-02j -4.0451e-02+2.9389e-02j  4.7553e-02+1.5451e-02j -1.5451e-02-4.7553e-02j -2.9389e-02+4.0451e-02j]
 [ 5.0000e-02-0.0000e+00j -1.5451e-02-4.7553e-02j -4.0451e-02+2.9389e-02j  4.0451e-02+2.9389e-02j  1.5451e-02-4.7553e-02j -5.0000e-02+4.2863e-17j  1.5451e-02+4.7553e-02j
   4.0451e-02-2.9389e-02j -4.0451e-02-2.9389e-02j -1.5451e-02+4.7553e-02j  5.0000e-02-8.5725e-17j -1.5451e-02-4.7553e-02j -4.0451e-02+2.9389e-02j  4.0451e-02+2.9389e-02j
   1.5451e-02-4.7553e-02j -5.0000e-02-4.9048e-17j  1.5451e-02+4.7553e-02j  4.0451e-02-2.9389e-02j -4.0451e-02-2.9389e-02j -1.5451e-02+4.7553e-02j]
 [ 5.0000e-02-0.0000e+00j -9.1849e-18-5.0000e-02j -5.0000e-02+1.8370e-17j -6.1263e-17+5.0000e-02j  5.0000e-02-3.6739e-17j  4.2894e-17-5.0000e-02j -5.0000e-02-1.2253e-16j
  -2.4524e-17+5.0000e-02j  5.0000e-02-7.3479e-17j -1.7148e-16-5.0000e-02j -5.0000e-02-8.5787e-17j  3.6749e-16+5.0000e-02j  5.0000e-02+2.4505e-16j -2.0822e-16-5.0000e-02j
  -5.0000e-02-4.9048e-17j  4.8955e-17+5.0000e-02j  5.0000e-02-1.4696e-16j  4.6558e-16-5.0000e-02j -5.0000e-02+3.4296e-16j -2.6958e-16+5.0000e-02j]
 [ 5.0000e-02-0.0000e+00j  1.5451e-02-4.7553e-02j -4.0451e-02-2.9389e-02j -4.0451e-02+2.9389e-02j  1.5451e-02+4.7553e-02j  5.0000e-02-4.8986e-17j  1.5451e-02-4.7553e-02j
  -4.0451e-02-2.9389e-02j -4.0451e-02+2.9389e-02j  1.5451e-02+4.7553e-02j  5.0000e-02-9.7972e-17j  1.5451e-02-4.7553e-02j -4.0451e-02-2.9389e-02j -4.0451e-02+2.9389e-02j
   1.5451e-02+4.7553e-02j  5.0000e-02-1.4696e-16j  1.5451e-02-4.7553e-02j -4.0451e-02-2.9389e-02j -4.0451e-02+2.9389e-02j  1.5451e-02+4.7553e-02j]
 [ 5.0000e-02-0.0000e+00j  2.9389e-02-4.0451e-02j -1.5451e-02-4.7553e-02j -4.7553e-02-1.5451e-02j -4.0451e-02+2.9389e-02j -3.6770e-17+5.0000e-02j  4.0451e-02+2.9389e-02j
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   4.0451e-02-2.9389e-02j  4.6558e-16-5.0000e-02j -4.0451e-02-2.9389e-02j -4.7553e-02+1.5451e-02j -1.5451e-02+4.7553e-02j  2.9389e-02+4.0451e-02j]
 [ 5.0000e-02-0.0000e+00j  4.0451e-02-2.9389e-02j  1.5451e-02-4.7553e-02j -1.5451e-02-4.7553e-02j -4.0451e-02-2.9389e-02j -5.0000e-02-1.2253e-16j -4.0451e-02+2.9389e-02j
  -1.5451e-02+4.7553e-02j  1.5451e-02+4.7553e-02j  4.0451e-02+2.9389e-02j  5.0000e-02+2.4505e-16j  4.0451e-02-2.9389e-02j  1.5451e-02-4.7553e-02j -1.5451e-02-4.7553e-02j
  -4.0451e-02-2.9389e-02j -5.0000e-02+3.4296e-16j -4.0451e-02+2.9389e-02j -1.5451e-02+4.7553e-02j  1.5451e-02+4.7553e-02j  4.0451e-02+2.9389e-02j]
 [ 5.0000e-02-0.0000e+00j  4.7553e-02-1.5451e-02j  4.0451e-02-2.9389e-02j  2.9389e-02-4.0451e-02j  1.5451e-02-4.7553e-02j  3.0647e-17-5.0000e-02j -1.5451e-02-4.7553e-02j
  -2.9389e-02-4.0451e-02j -4.0451e-02-2.9389e-02j -4.7553e-02-1.5451e-02j -5.0000e-02-6.1294e-17j -4.7553e-02+1.5451e-02j -4.0451e-02+2.9389e-02j -2.9389e-02+4.0451e-02j
  -1.5451e-02+4.7553e-02j -2.6958e-16+5.0000e-02j  1.5451e-02+4.7553e-02j  2.9389e-02+4.0451e-02j  4.0451e-02+2.9389e-02j  4.7553e-02+1.5451e-02j]]
x =
[ 1.15  +0.0000e+00j  0.7069+6.6002e-01j  0.0345+5.0921e-01j -0.0075-9.1499e-05j  0.3095-6.5716e-02j  0.3   +1.5000e-01j  0.0905+7.9592e-02j  0.1603-1.4540e-01j
  0.3655-1.0633e-01j  0.3403+4.4486e-02j  0.25  -3.1241e-16j  0.3403-4.4486e-02j  0.3655+1.0633e-01j  0.1603+1.4540e-01j  0.0905-7.9592e-02j  0.3   -1.5000e-01j
  0.3095+6.5716e-02j -0.0075+9.1499e-05j  0.0345-5.0921e-01j  0.7069-6.6002e-01j]
```
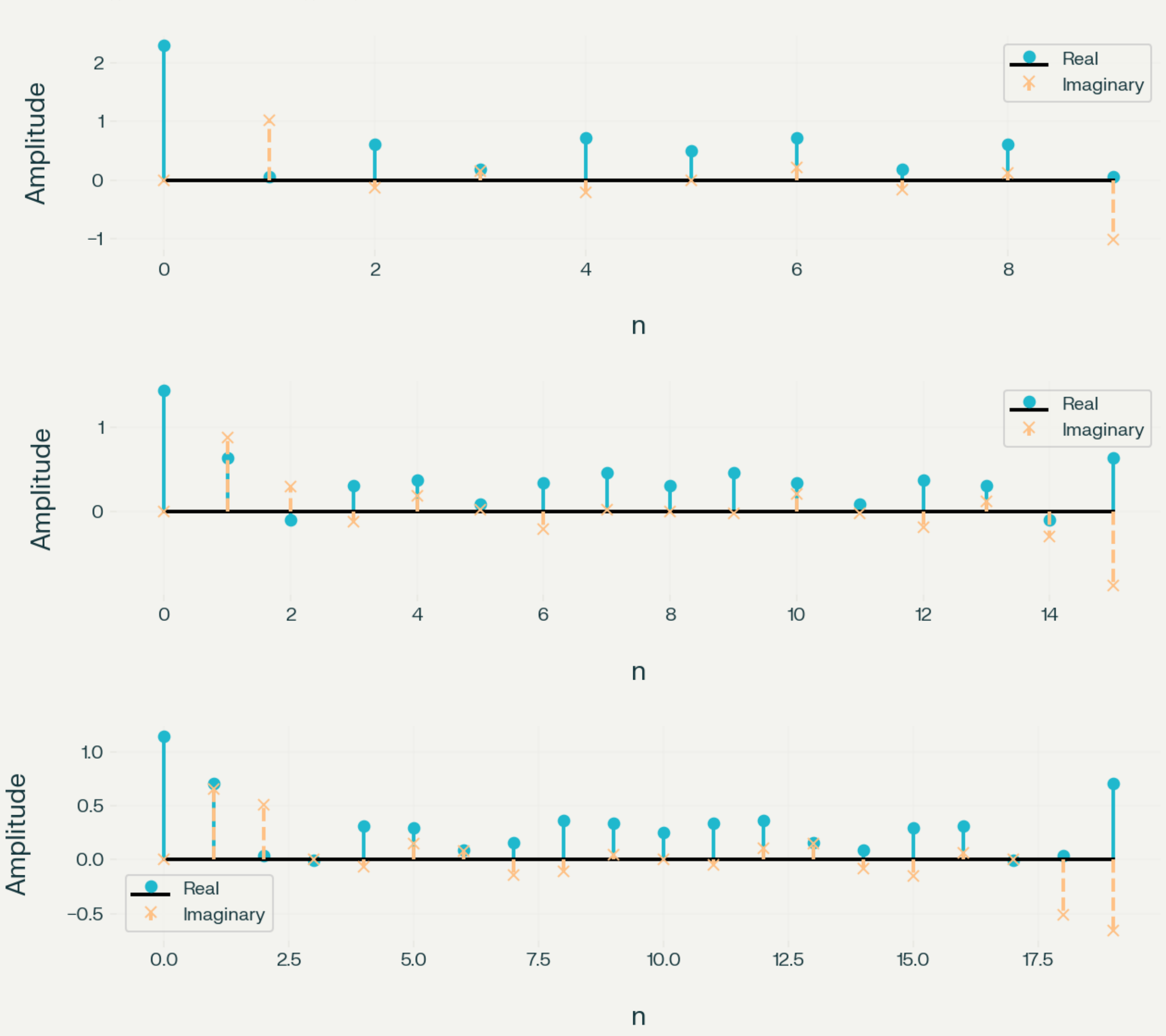

Executed output: full W, K and synthesized samples

```
/usr/local/lib/python3.12/site-packages/pydantic/main.py:463: UserWarning: Pydantic serializer warnings:
  PydanticSerializationUnexpectedValue(Expected `LineChart` - serialized value may not be as expected [input_value=Chart(type=<ChartType.UNK...e='N = 10', elements=[]), input_t
  PydanticSerializationUnexpectedValue(Expected `ScatterChart` - serialized value may not be as expected [input_value=Chart(type=<ChartType.UNK...e='N = 10', elements=[]), inpu
  PydanticSerializationUnexpectedValue(Expected `BarChart` - serialized value may not be as expected [input_value=Chart(type=<ChartType.UNK...e='N = 10', elements=[]), input_ty
  PydanticSerializationUnexpectedValue(Expected `PieChart` - serialized value may not be as expected [input_value=Chart(type=<ChartType.UNK...e='N = 10', elements=[]), input_ty
  PydanticSerializationUnexpectedValue(Expected `BoxAndWhiskerChart` - serialized value may not be as expected [input_value=Chart(type=<ChartType.UNK...e='N = 10', elements=[])
  PydanticSerializationUnexpectedValue(Expected `LineChart` - serialized value may not be as expected [input_value=Chart(type=<ChartType.UNK...e='N = 16', elements=[]), input_t
  PydanticSerializationUnexpectedValue(Expected `ScatterChart` - serialized value may not be as expected [input_value=Chart(type=<ChartType.UNK...e='N = 16', elements=[]), inpu
  PydanticSerializationUnexpectedValue(Expected `BarChart` - serialized value may not be as expected [input_value=Chart(type=<ChartType.UNK...e='N = 16', elements=[]), input_ty
  PydanticSerializationUnexpectedValue(Expected `PieChart` - serialized value may not be as expected [input_value=Chart(type=<ChartType.UNK...e='N = 16', elements=[]), input_ty
  PydanticSerializationUnexpectedValue(Expected `BoxAndWhiskerChart` - serialized value may not be as expected [input_value=Chart(type=<ChartType.UNK...e='N = 16', elements=[])
  PydanticSerializationUnexpectedValue(Expected `LineChart` - serialized value may not be as expected [input_value=Chart(type=<ChartType.UNK...e='N = 20', elements=[]), input_t
  PydanticSerializationUnexpectedValue(Expected `ScatterChart` - serialized value may not be as expected [input_value=Chart(type=<ChartType.UNK...e='N = 20', elements=[]), inpu
  PydanticSerializationUnexpectedValue(Expected `BarChart` - serialized value may not be as expected [input_value=Chart(type=<ChartType.UNK...e='N = 20', elements=[]), input_ty
  PydanticSerializationUnexpectedValue(Expected `PieChart` - serialized value may not be as expected [input_value=Chart(type=<ChartType.UNK...e='N = 20', elements=[]), input_ty
  PydanticSerializationUnexpectedValue(Expected `BoxAndWhiskerChart` - serialized value may not be as expected [input_value=Chart(type=<ChartType.UNK...e='N = 20', elements=[])
  return self.__pydantic_serializer__.to_python(
```



Interpretation
The output is complex because the input spectrum is not conjugate symmetric. At n = 0 the values are 23/N: 2.3, 1.4375 and 1.15 for N = 10, 16 and 20. Keeping the coefficients fixed while increasing N changes their normalized frequencies and the 1/N scale; this is not amplitude-preserving interpolation of an unchanged waveform.

Reference
Brian McFee, Digital Signals Theory, section 7.2, The Inverse DFT.
https://brianmcfee.net/dstbook-site/content/ch07-inverse-dft/InverseDFT.html